

ME 780: Vehicle System Dynamics

Assignment #3

Due date: August 03, 2026

Please use the course project template for your assignment report.

There should not be any changes to the template format.

Description:

Consider a **hatchback** with the following specifications:

Vehicle curb weight= 1,000kg

Total vehicle sprung mass weight = 900kg

Unsprung mass weight per wheel (front and rear) = 25kg

Vehicle inertial about z axis = 2000kg.m²

Sprung mass roll inertia =750kg.m²

Wheel base = 2.6m

Track (front and rear) =1.4m

Distance of CG from front axle = 1.2 m

Distance of CG from rear axle = 1.4 m

Vehicle CG height =0.6 m

Unsprung mass CG height = 0.7m

Roll center height (front/rear) = 0.2m

Tire cornering stiffness (front and rear) = 25,000 N/rad

Tire vertical stiffness = 150,000 N/m

Front shock absorber rate (per wheel) = 800 N/m/s

Rear shock absorber rate (per wheel) = 1,000 kg/s

Distance between installation point of Left and right springs (front and rear) = 1.3 m

Distance between installation point of Left and right shock absorber (front and rear) = 1.3 m

Installation factor for springs and shock absorbers for front and rear wheels = 1

Part 1: Ride Comfort Analysis:

- 1- For the above vehicle, calculate the front and rear spring rates to have:
 - a) The first natural frequency of the front suspension to be 1.15Hz
 - b) The first natural frequency of the rear suspension to be 1.25Hz
- 2- Derive the 1-DOF and 2-DOF front quarter model equations of the above vehicle and:
 - a) Plot the sprung mass displacement and acceleration transmissibility functions by using 1-DOF quarter car model, discuss the results.

- b) Simulate the vehicle response to a half-sine road excitation (magnitude = 8 cm, wavelength = 1.0m) by using the 1-DOF and 2-DOF quarter car models for 10km/h and 50km/h vehicle speeds. Compare and discuss the sprung and unsprung mass displacement, acceleration, and wheel travel results.

Part 2: Handling Analysis, bicycle model

- 1- Calculate the understeer coefficient of the above vehicle and discuss the vehicle understeering/oversteering characteristics.
- 2- Find the analytical form of the yaw velocity gain and plot it versus vehicle speed.

Simulation

Consider a step steer maneuver (wheel steering angle = 2.0 deg) at 20m/s vehicle speed and simulate the vehicle handling responses for the bicycle model and plot yaw rate, lateral velocity, lateral acceleration, vehicle path, vehicle side slip angle, and front and rear tires side slip angles.

Part 3: Vehicle Control

Pick one vehicle from CarSim (e.g. E-class SUV). Using the vehicle's parameters in the CarSim model, find the steady state yaw rate (r_{ss}) as a function of vehicle speed and steering angle.

For this vehicle and the steady state yaw rate (r_{ss}) found above:

- a) Design a feedback controller (closed-loop control system) of your choice to track the desired yaw rate. The steady state yaw rate (r_{ss}) is considered as the desired yaw rate.
- b) Simulate the response of the vehicle in CarSim for a constant velocity of 60 kph and steering angle of $\delta_w = 90 \sin\left(\frac{\pi}{2}t\right)$ for two road friction coefficients of 0.55 and 0.90. Compare and discuss the results when the controller is off and on. You are allowed to make any assumptions needed for controlling the vehicle.

Hint: you should connect CarSim to Matlab/Simulink and create the control loop as below:



Your control signal should be ΔM (extra moment at vehicle CG) that generates by differential braking.

- c) Discuss the results and any differences between the vehicle responses on the two roads with different friction coefficients.